Deployment Document | Aryan Parmar, Raad Islam, Marko Simic, Tobi Ajadi

1. Hosting Infrastructure

The application is deployed on a Microsoft Azure Virtual Machine. The Virtual Machine hosts the fully configured Java environment and the compiled application files.

The database is hosted externally using Neon, a PostgreSQL cloud database service. The application connects to the Neon database instance using secure credentials configured in a .env file.

2. Application Deployment on Azure Virtual Machine

The following steps were completed to host the application on Azure Virtual Machine:

1. Completed Azure setup of Virtual Machine Instance selecting the VM ram, disk space ,and Operating System.
2. Installing XFCE and RDP for the VM
3. Used Microsoft Remote Desktop app to connect to the VM using appropriate credentials for login
4. Installed NoMachine on VM to allow connections from NoMachine by users
5. Downloaded JDK and required dependencies
6. Cloned repository using git clone https://github.com/notaarryan/Comp2800Project.git
7. Created .env using echo "DATABASE_URL=neaon_connection_string" > .env
8. Created a StartWhiteboard.sh file on desktop

StartWhiteboard.sh contains the following bash script to ensure latest version is displayed to the user:

```
#!/bin/bash
cd /home/COMP2800gr5/Desktop/Comp2800Project git pull javac -d
out/production/Comp2800Project src/*.java && java -cp
out/production/Comp2800Project:lib/* whiteboardGUI
```

The script was made executable using:

```
chmod +x ~/Desktop/StartWhiteboard.sh
```

3. How Graders can access the Application on the VM

Graders can access the deployed application by connecting to the Azure VM using Remote Desktop Protocol (RDP).

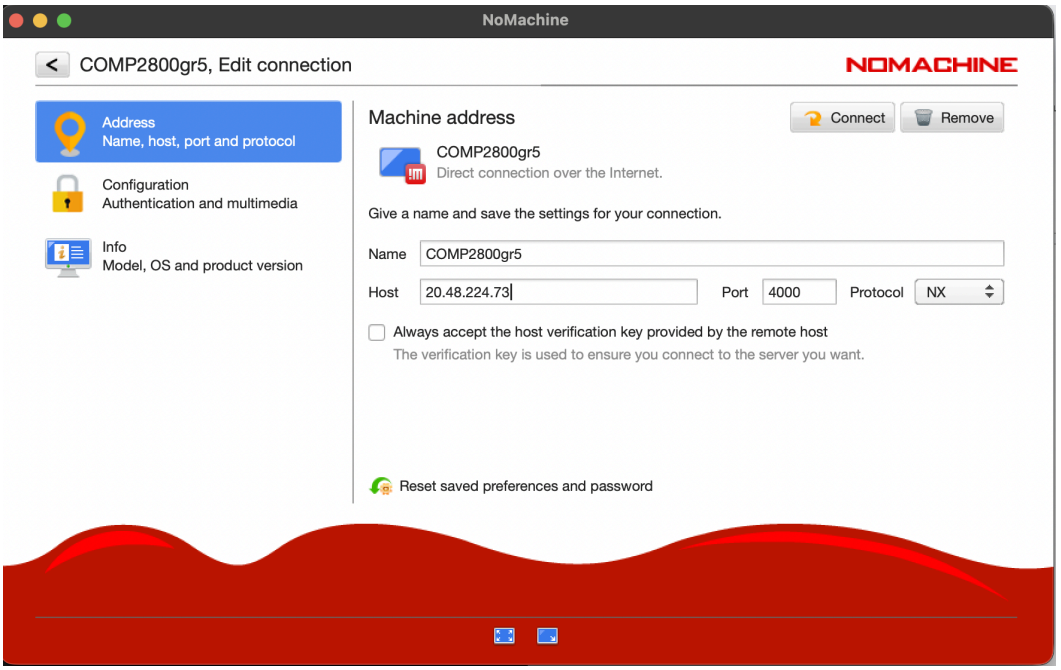
The VM can be accessed using any RDP client, including:

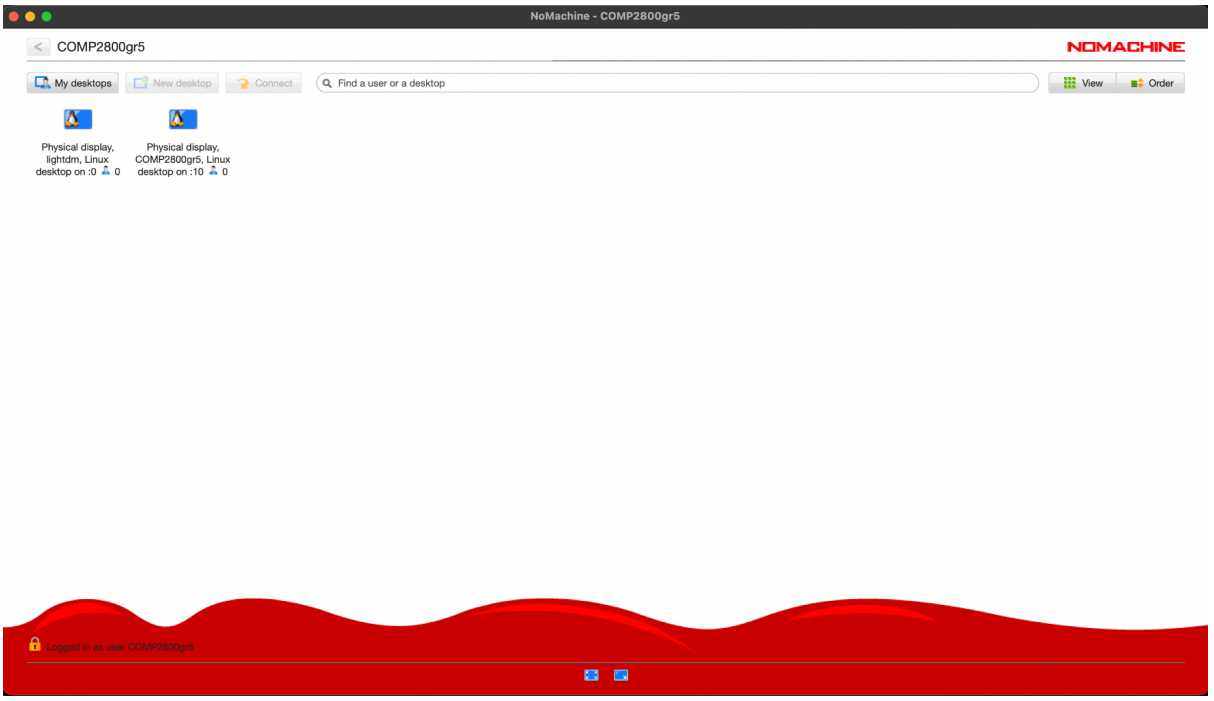
- Windows Remote Desktop Connection (mstsc)
- Microsoft Remote Desktop application
- NoMachine

4. Connection Information

VM Public IP: 20.48.224.73
Username: COMP2800gr5
Password: COMP2800gr5@Whiteboard

Steps to connect on NoMachine:

1. 

2. Enter Username and Password specified before
3. 

Select Physical display, COMP2800gr5, Linux desktop on:10

5. Database Deployment

The database was deployed on neon which required the following steps:

1. Creating Neon project for the Whiteboard Database
2. Creating a database schema (schema.sql)
3. Running schema.sql within neon sql editor to create the required tables
4. Get the connection string from dashboard

The repository includes schema.sql but it is not to be ran by graders, as the database is already setup on Neon

For grades to connect to the Neon database follow these steps:

1. Create a .env outside of src directory
2. Write this into the .env: DATABASE_URL=jdbc:postgresql://ep-dark-frog-an5w50xn-pooler.c-6.us-east-1.aws.neon.tech/whiteboard?user=neondb_owner&password=npg_URp0YxgO2bqw&sslmode=require&channelBinding=require
3. Everything else is handled by the Application

6. Repository Link

The complete source code, database schema, and documentation are available in the project repository:
https://github.com/notaarryan/Comp2800Project/tree/feature/collaborative-whiteboard